



The American University in Cairo

School of Sciences and Engineering

Department of Computer Science and Engineering

CSCE 345 / 3401 – Operating Systems

Spring 2026

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Semester Project: Linux Network Monitor & Controller

1. Project Vision

The objective of this project is to design, implement, and evaluate a comprehensive Linux Network Monitoring and Control System.

Your tool must:

- Visually represent real-time network traffic flows
- Correlate traffic with individual processes and users
- Provide meaningful insights into system-level behavior
- Demonstrate strong systems-level understanding and engineering rigor

This project emphasizes systems thinking, design maturity, and technical depth. For examples of existing tools, see [1](#), [2](#), and [3](#).

2. Phase I – Survey, Analysis & Requirements (30%)

A. Tool Survey & Comparative Study

You are required to:

- Identify and analyze leading Linux network monitoring tools
- Evaluate architectures, strengths, and limitations
- Assess usability, performance, scalability, and extensibility
- Extract lessons learned to guide your design decisions

B. Requirements Specification

Define the following:

- Functional requirements
- Non-functional requirements (performance, reliability, scalability, security)
- System constraints and architectural assumptions
- User interaction model and GUI expectations

Phase I Deliverable:

- Due: March 25 (In-Class)
- Includes: Survey report + Requirements specification
- Presentation: Group presentation
- Weight: 30% of project grade

3. Phase II – Design, Implementation & Demonstration (50%)

Primary Language: C++

Bonus: Rust implementation earns +5% bonus credit

Your implementation must demonstrate:

- Accurate packet monitoring

- Process and user-level traffic correlation
- Efficient system-level resource management
- Clean, modular, and maintainable architecture

Graphical User Interface (GUI)

Your GUI must include:

- Clear real-time traffic visualization
- Process-user mapping representation
- User-friendly controls and intuitive navigation

Phase II Deliverable:

- Due: May 10 (In-Class)
- Includes: Final system + technical report + live demo
- Weight: 50% of project grade

4. Checkpoints (20%)

- April 1 – 10%
- April 19 – 10%

Each checkpoint requires meaningful progress demonstration, architectural clarity, and prototype validation.

5. Evaluation Criteria

Projects will be evaluated based on:

- Design quality and architectural soundness
- Efficiency and performance
- Innovation and robustness
- Documentation clarity and technical depth
- Quality of live demonstration

Grades are earned through engineering excellence and demonstrated systems-level mastery.